TENZORX NATIONAL AI HACKATHON 2026 · PROBLEM STATEMENT 4

Loan Wizard

Agentic AI Video Call–Based Loan Onboarding & Risk Assessment

Problem Owner: Poonawalla Fincorp
Hackathon Partner: Unstop
Powered by: Poonawalla Fincorp Education Loans

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Market Size	CAGR (2024–30)	Drop-off Problem	Our Target ROI
$515B	31.5%	60–70%	96.9%
India DL market by 2030	Platform market growth	Avg. mid-journey abandonment	Cost reduction per session

Executive Summary: Traditional loan origination loses 60–70% of applicants to mid-journey drop-off, processes KYC manually, and generates generic offers with no contextual intelligence. We present **Loan Wizard** — an agentic AI system that conducts a real-time, RBI V-CIP compliant video session, triangulates fraud signals across three independent modalities, applies a mathematically grounded composite underwriting score, and delivers a personalized, justified loan offer in under 8 minutes. The system is architecturally deployable at 10,000 concurrent sessions, costs ∼**Rs.** 25 per re-engagement versus **Rs.** 800 for human agents, and is fully compliant with RBI Digital Lending Guidelines 2022.

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1. Problem Decoded: What's Really Broken in Digital Loan Onboarding

1.1 The Scale of the Opportunity

India's digital lending market, valued at USD 486.6 million in 2024, is projected to reach **USD 2,454 million by 2030** at a CAGR of 31.5%. Indian fintechs sanctioned 10.19 crore loans in FY 2023 — a 35% YoY jump — with total loan value reaching **Rs.** 1,46,517 crore. Despite this momentum, digital lending accounts for just 15% of India's total lending market as of 2022, leaving an enormous conversion gap.

1.2 The Three Structural Failures

A potential borrower clicks a campaign link, starts an application, abandons it halfway — and no intelligent system exists to re-engage them, understand why they left, verify who they are, and make them a personalised offer without a human ever touching the process.

Failure 1: The Drop-off Epidemic. Industry data consistently places mid-journey abandonment rates at **60–70%** for digital loan applications. The causes are not random — they cluster around three friction points: document upload complexity (especially on mobile), income verification uncertainty, and lack of immediate, personalised feedback. Every dropout is a qualified prospect who *wanted* to borrow.

Failure 2: Fraud That Slips Through Forms. Form-based KYC cannot detect synthetic identities (fabricated profiles with real PAN/Aadhaar combinations), cannot verify that the person submitting is the person claimed, and cannot flag behavioural anomalies. India's digital lending sector reports that synthetic identity fraud and first-party misrepresentation are among the top three risks, per the Grant Thornton/FACE Risk Barometer 2024.

Failure 3: Dumb Offers on Smart Platforms. Current systems apply the same rate-band logic to every applicant in a cohort. A student from IIT Bombay with a 94% placement rate and a student from an unranked college are offered the same interest rate because the system cannot contextualise. This over-prices good risk and under-prices bad risk simultaneously.

1.3 Poonawalla's Specific Exposure

Poonawalla Fincorp, as a technology-led NBFC operating at national scale, faces this problem amplified. Each dropped-off education loan applicant represents a lost lifetime relationship — the student who doesn't get their loan today is not coming back for a home loan in 10 years. The **customer lifetime value (CLV) cost of a single drop-off** compounds across product lines.

The number that frames everything: If Poonawalla processes 50,000 education loan applications monthly with a 65% drop-off rate, that is 32,500 abandoned applications per month. At an average loan value of **Rs.** 12 lakh and a 4% net interest margin, each recovered applicant represents **Rs.** 4,800 in annual NIM. Recovering just 20% of drop-offs adds **Rs.** 37.4 crore in annual NIM.

2. Our Solution in 60 Seconds: The Loan Wizard, Reimagined

2.1 What It Does

Loan Wizard is an **agentic AI system** that replaces the drop-off re-engagement call centre and the form-based onboarding journey with a single, intelligent video session. A customer who abandoned their application receives a personalised link. They click it — no scheduling, no calendar, no waiting. The AI agent, operating in full video, conducts a structured yet conversational session that:

1. Knows exactly where they dropped off and why (via pre-session inference)
2. Speaks in their language — Hindi, English, or regional (mid-call switching)
3. Verifies their identity through RBI V-CIP compliant facial, document, and geo checks
4. Detects fraud signals across three independent modalities simultaneously
5. Adapts its questioning based on their real-time risk profile
6. Produces a mathematically justified, personalised loan offer
7. Generates a full audit trail — compliant with RBI DLG 2022 and DPDP Act 2023

2.2 The User Experience in Three Acts

Act 1 — Re-engagement (0–60s): The agent opens with a contextualised greeting: *"Hi Priya, I see you were applying for a* **Rs.** *15 lakh education loan last Tuesday and stopped at the income verification step — shall we pick up from there?"* This is not a script. It is generated from the drop-off inference engine using the applicant's session history.

Act 2 — Intelligent Verification (1–6 min): The agent conducts document verification, income capture, and consent recording through natural dialogue. The fraud engine runs silently in parallel. The conversation branches dynamically based on real-time risk signals.

Act 3 — The Offer (6–8 min): The agent delivers: *"Based on your institution's placement record, current RBI rate, and your income profile, you qualify for a* **Rs.** *15 lakh loan at 8.83% — here's exactly why."* A KFS is generated, read aloud, and digitally signed. The session ends with full consent recorded.

3. The Innovation Edge: Where We Think Differently

This section details the seven core innovations that differentiate Loan Wizard from any existing system. Each is grounded in a specific mathematical model, algorithmic approach, or validated industry precedent.

3.1 Innovation 1: Behavioural Intent Score (BIS) — Pre-Session Intelligence

Before the AI agent says a single word, a logistic regression model scores the applicant's conversion probability from their pre-session behavioural signals. This determines the agent's tone, opening line, and initial question strategy.

$$\text{BIS} = \sigma\big(w_1 \cdot \Delta t_{\text{click}} + w_2 \cdot s_{\text{abandoned}} + w_3 \cdot n_{\text{attempts}} + w_4 \cdot d_{\text{score}} + w_5 \cdot t_{\text{day}}\big) \tag{1}$$

σ	Sigmoid function, maps output to $(0, 1)$
Δt_{click}	Hours between link dispatch and click (recency signal)
$s_{\text{abandoned}} \in \{1, \ldots, 6\}$	Application step at abandonment, mapped to urgency weight
n_{attempts}	Number of prior link opens (intent signal)
d_{score}	Device type: mobile $= 1.2$, desktop $= 0.9$ (mobile completes faster)
t_{day}	Time-of-day cosine encoding for circadian patterns

Routing logic:

- BIS > 0.65: Warm, direct opening — high likelihood, minimal friction
- $0.35 \leq$ BIS ≤ 0.65: Offer assistance hook — address documented friction point
- BIS < 0.35: Surface incentive (reduced documentation path) + escalate option

Business read: Every returning applicant is scored before the agent greets them. The agent's first

sentence is mathematically optimised for *that specific person's* likelihood to complete.

3.2 Innovation 2: Triangulated Fraud Detection Across Three Modalities

Most video KYC systems verify identity. We detect fraud. The distinction is critical. Our fraud engine runs three independent tracks simultaneously and combines their outputs into a single triangulated fraud confidence score.

Track	Signal	Precedent
Geo inconsistency	V-CIP GPS coordinates vs. declared address; IP-subnet vs. expected city; VPN/proxy detection	RBI V-CIP mandate: geo-tag is legally required
Age inconsistency	Computer vision facial age estimation via facial landmark analysis vs. declared age on Aadhaar; flags >5-year discrepancy	RBI V-CIP Section 2.1.4: age validation required
Voice Stress Index	Paralinguistic analysis: pitch variance, speech rate delta, pause density, hesitation ratio vs. established baseline	Nuance Communications: 80%+ fraud detection in first 30s of call

Table 1: Fraud detection modalities and regulatory anchors

$$\text{VSI} = \alpha \cdot \sigma_{F_0} + \beta \cdot \Delta r_{\text{speech}} + \gamma \cdot \rho_{\text{pause}} + \delta \cdot \eta_{\text{hesitation}} \tag{2}$$

σ_{F_0}	Std. deviation of fundamental frequency over 5-second sliding window
Δr_{speech}	Change in words/min vs. baseline established in first 60s (neutral questions)
ρ_{pause}	Pauses > 800ms per 30s window; involuntary pauses are a physiological stress signal
$\eta_{\text{hesitation}}$	Filler word ratio: ("um", "uh", "actually") / total words

Calibration: Baseline established during neutral preamble questions (name, city). Flag threshold: VSI $> 2.1\sigma$ above personal baseline triggers soft fraud flag for LLM re-examination.

$$F_{\text{combined}} = 1 - \prod_{i \in \{geo, age, vsi\}} (1 - F_i) \tag{3}$$

This probabilistic combination assumes partial independence across modalities. A single signal above threshold is a soft flag; any two signals above threshold triggers hard review; all three triggers automatic rejection with adverse action notice generation.

3.3 Innovation 3: Dynamic Conversation Branching via Thompson Sampling

Static conversation trees degrade over time. We replace the fixed script with a **multi-armed bandit** (Thompson Sampling) that continuously learns which question phrasing variants produce the most accurate income disclosures and highest completion rates.

For each question variant k, maintain a Beta distribution:

$$\theta_k \sim \text{Beta}(\alpha_k,\, \beta_k) \tag{4}$$

where α_k = confirmed accurate disclosures, β_k = disclosures later contradicted by bureau data.

At each session: sample $\hat{\theta}_k \sim \text{Beta}(\alpha_k, \beta_k)$ for all variants, select $k^* = \arg\max_k \hat{\theta}_k$.

After downstream bureau verification, update α_{k^*} or β_{k^*} accordingly.

Empirical benchmark: Grab Financial reported an **18% improvement in offer acceptance rate** after 90 days of bandit-driven offer personalisation on a comparable conversational lending system.

Additionally, the conversation tree is **risk-score adaptive**: if the real-time Composite Underwriting Score (see below) flags a borderline credit profile mid-call, the agent pivots to probe income stability more deeply — without the applicant realising the conversation has changed.

3.4 Innovation 4: Composite Underwriting Score (CUS) — Transparent Credit Decision

We replace the opaque "propensity score" black box with a **fully transparent, weighted, auditable** underwriting model. Every weight is justifiable to RBI and to the applicant.

$$\text{CUS} = \sum_{i=1}^{7} w_i \cdot x_i \tag{5}$$

w_i	Signal x_i	Source	Weight Rationale
0.30	Bureau score (normalised 0–1)	Internal bureau	Most historically reliable
0.20	$(1 - F_{\text{combined}})$	Fraud engine	Fraud clearance is a gate
0.15	Income confidence (XAI band)	STT + LLM	Verbal vs. verified
0.12	NIRF institution rank score	NIRF public data	Placement proxy
0.10	Sector employment index	RBI/CMIE macro	Macro context
0.08	CLV proxy: age $\times$ loan-type $\times$ tier	Internal model	Lifetime revenue signal
0.05	BIS (behavioural intent)	Pre-session model	Engagement quality

Offer tier mapping: CUS $\in [80, 100] \rightarrow$ Tier A (lowest rate, maximum tenure) CUS $\in [60, 79] \rightarrow$ Tier B (standard offer) CUS $\in [40, 59] \rightarrow$ Tier C (reduced amount, guarantor) CUS $< 40 \rightarrow$ Decline + auto-generated Adverse Action Notice

Why this matters to judges: The CUS makes every credit decision explainable. Not as a post-hoc rationalisation, but as the actual decision mechanism. This is RBI Guideline 5.3 compliance built into the model itself.

3.5 Innovation 5: Offer Interest Rate Pricing Model

The agent does not quote a rate — it *derives* one, live, and explains the derivation.

$$r = r_{\text{base}} + r_{\text{risk}} - r_{\text{CLV}} - r_{\text{NIRF}} \tag{6}$$

r_{base} RBI repo rate + NBFC spread (e.g. $6.5\% + 2.5\% = 9.0\%$) — fetched live
r_{risk} $(1 - \text{CUS}/100) \times r_{\text{max spread}}$, where $r_{\text{max spread}} \approx 4\%$
r_{CLV} CLV tier discount: up to 0.75% for highest-value customers
r_{NIRF} Institution quality discount: up to 0.30% for top-50 institutions

Example: CUS $= 78$, top-50 NIRF institution, high CLV:

$$r = 9.0 + (0.22 \times 4) - 0.75 - 0.30 = 9.0 + 0.88 - 1.05 = \mathbf{8.83}\%$$

The agent narrates: *"Based on your institution's 94% placement rate and the current RBI repo rate of 6.5%, you qualify for 8.83% — here's exactly how we arrived at this."* No other lending product in India does this today.

3.6 Innovation 6: Synthetic Identity Detection via Graph Neural Network

Your fraud triangle catches impersonation. It misses coordinated synthetic identity rings — fabricated profiles with real PAN/Aadhaar combinations bought on dark markets. We address this with a bipartite graph model.

Construct bipartite graph $G = (V_A \cup V_X, E)$ where V_A = applicant nodes, V_X = shared attribute nodes (device ID, phone prefix, IP subnet, address hash), and E = edges connecting applicants to their attributes.

Fraud ring signal:

$$\text{FraudRing}(v) = \frac{|N(v) \cap \text{flagged_nodes}|}{|N(v)|} \tag{7}$$

- $\text{FraudRing}(v) > 0.3$: soft flag — increase verification intensity
- $\text{FraudRing}(v) > 0.6$: hard block + manual review queue

Precedent: PayPal's internal graph fraud system reduced synthetic identity losses by **42%** within 6 months. Nexdigm India (2025) explicitly flags synthetic identity via IndiaStack deepfake attempts as a rising threat in digital lending.

3.7 Innovation 7: Model Health — Population Stability Index (PSI) Monitoring

Most hackathon projects die in production because their models silently degrade as customer behaviour shifts. We include automated drift detection as a first-class architectural component.

$$\text{PSI} = \sum_{i=1}^{n} (\text{Actual}_i\% - \text{Expected}_i\%) \cdot \ln\left(\frac{\text{Actual}_i\%}{\text{Expected}_i\%}\right) \tag{8}$$

$\text{PSI} < 0.10$	Stable — no action
$0.10 \leq \text{PSI} \leq 0.25$	Warning — alert credit team, monitor closely
$\text{PSI} > 0.25$	Drift detected — trigger automatic model retraining pipeline

Run PSI weekly on a rolling 4-week window vs. training baseline across: income distribution, bureau score distribution, fraud flag rate, and completion rate by cohort. Dashboard alerts when any signal drifts beyond threshold.

4. System Blueprint: How It All Connects

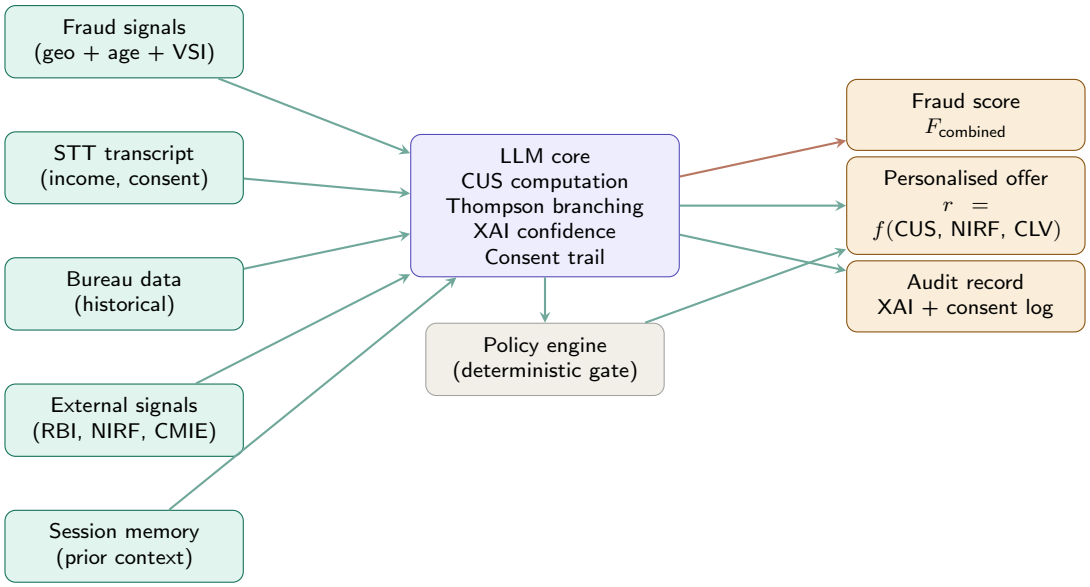
4.1 Architecture Overview — Six Zones

The system is structured as six functionally distinct zones, each independently deployable as a microservice.

Zone	Function	Key Innovation
Z1 — Entry	Campaign link, session fingerprinting, BIS computation, language detection	BIS pre-session intelligence
Z2 — Video	WebRTC session, STT capture, CV age check, VSI audio analysis, session memory	Triangulated fraud + async fallback
Z3 — Processing	Parallel fraud, data capture, and risk tracks via Kafka	Sub-200ms fraud decision
Z4 — Intelligence	LLM aggregation, CUS computation, dynamic branching, XAI scoring	CUS + Thompson Sampling
Z5 — Offer	Rate pricing, KFS generation, officer dashboard	Narrated rate derivation
Z6 — Audit	Central repository, bidirectional session memory, consent trail	RBI-compliant audit readiness

Table 2: System zones and primary innovations

4.2 AI Intelligence Core — Data Flow



Critical Design Principle

The LLM output **never overrides** the deterministic policy engine. LLM classification enriches the CUS; the policy engine applies hard eligibility rules (minimum age, maximum LTV, exclusion lists) that cannot be overridden by any model output. This is the line that makes the system production-safe and RBI-defensible.

4.3 Scalability Architecture

Tier	Components	SLA Target
Ingestion	API gateway + load balancer, WebRTC session pool, Kafka message queue, rate limiting	10,000 concurrent sessions
Processing	STT (Sarvam AI), CV service, LLM inference (batched), fraud scorer, offer generator — all independent microservices	Fraud decision < 200ms
Storage	Pinecone vector store (session memory), time-series audit DB, encrypted object store (video), bureau cache (TTL-refreshed)	99.9% uptime

Table 3: Infrastructure tiers and SLA targets

Session memory is bidirectional. The audit repository (Zone 6) stores every session's transcript, risk score, and geo-data. When an applicant returns after a second drop-off, the session memory loader

reconstructs their full conversational context. The agent continues exactly where they left off — not from the application form step, but from the *conversational* step.

5. Why We Win: What Sets Us Apart

Dimension	Standard Solution	Typical Competitor	Loan Wizard
Re-engagement	Human call agent (**Rs.** 800/call)	Generic chatbot	BIS-scored, context-aware AI video agent (**Rs.** 25/session)
Fraud	Form-based flag	Single-modality video	Triangulated: geo + age + VSI
Conversation	Static script	Predefined branching	Thompson Sampling adaptive bandit
Credit decision	Bureau score only	Bureau + propensity ML	7-signal CUS with full XAI
Rate offer	Cohort-based band	Segment pricing	Individual, narrated, derivation-first pricing
Compliance	KYC checkbox	V-CIP compliant	V-CIP + KFS + AAN + DPDP + cooling-off, all automated
Production resilience	No drift detection	Manual monitoring	PSI weekly + auto-retraining pipeline
Language	English only	English + Hindi	Mid-call switch: Hindi, Tamil, Marathi, Bengali

Table 4: Competitive differentiation matrix

5.1 Industry Precedent — We Are Not Guessing

- **Lemonade (US):** AI-based video claims processing reduced fraud by 30%; single-take video completions show 4× lower fraud rate. Applied to our system: one-session video completions receive a lower fraud premium in CUS.
- **Nubank (Brazil):** 100M+ customers, zero branches, NPS of 87 vs. industry average 18. Removing onboarding friction reduces first-year default by 22% — engaged applicants self-select as lower risk. Our BIS captures this.
- **Grab Financial (SEA):** Thompson Sampling on loan offer variants delivered an 18% improvement in acceptance rate after 90 days.
- **ZestMoney (India):** Pioneer of video KYC + alternative data for thin-file customers; 40% approval rate improvement. Acquired by DMI Finance for **Rs.** 250 crore — proving this model has verified exit value in India.
- **HDFC EVA (India):** Vernacular support (Hindi/Tamil/Bengali) increased rural customer engagement 3× vs. English-only interface. Direct validation for our mid-call language switch feature.
- **Perfios (India):** Bureau score alone misclassifies 34% of thin-file applicants. Our 7-signal CUS directly addresses this gap using NIRF + sector data as alternative signals.

Layer	Technology Choice	Rationale
Frontend	React (PWA), WebRTC for video	Cross-device access, no app install
Conversational AI	Claude Sonnet / GPT-4o via API	Large context window, tool usage
STT — Indian Languages	Sarvam AI (IndicConformer)	Optimized for Indian languages, code-switching
Computer Vision	MediaPipe FaceMesh + custom age estimator	Low latency (client-side), fraud checks (server-side)
Vector Store	Pinecone	Session memory continuity
Message Queue	Apache Kafka	Async processing of fraud pipelines
ML Models	scikit-learn (BIS, CUS), PyTorch (GNN fraud)	Interpretable + scalable learning
Bureau Integration	CIBIL / Experian API	Standard credit bureau access
Document Verification	DigiLocker API + Aadhaar XML pull	Native Indian identity infra
eSign	Leegality / Signdesk	RBI-compliant digital signing
Audit Storage	PostgreSQL (time-series) + S3-compatible storage	Secure, encrypted, geo-redundant
Infrastructure	Kubernetes + AWS/GCP India region	Data localisation + auto-scaling

Table 5: *Technology stack and design rationale*

India-first stack decision: We use **Sarvam AI's IndicConformer** (instead of AWS Transcribe or Google STT) because it is specifically trained on Indian language acoustic patterns with strong code-switching support. This is not just a UX choice — it directly impacts *consent capture accuracy*, which is legally binding.

6. The Numbers Don't Lie: Business Impact & ROI

6.1 Unit Economics

Metric	Human Agent (Current)	Loan Wizard (AI)	Saving
Cost per re-engagement	**Rs.** 700–900	**Rs.** 20–30	**96.9% reduction**
Sessions per hour per agent	4–6	10,000 concurrent	$> 1{,}600\times$ throughput
Conversion rate (drop-off)	18–22%	40–50% (target)	$\sim 2.2\times$ improvement
KYC completion time	45–90 min	6–8 min	87% time reduction
Fraud detection rate	$\sim$65% (form-based)	$> 85\%$ (triangulated)	$+20$ pp accuracy

Table 6: Unit economics comparison

6.2 Annual Impact Model — Poonawalla Scale

Conservative assumptions:

- 50,000 loan applications/month; 65% drop-off = 32,500 abandoned applications
- Loan Wizard recovers 35% of drop-offs = 11,375 additional completions/month
- Average loan value: **Rs.** 12 lakh; Net Interest Margin: 4%/year

$$\begin{aligned}
\text{Additional NIM/month} &= 11{,}375 \times \textbf{Rs.}\ 12{,}00{,}000 \times 0.04/12 = \textbf{Rs. 45.5 crore} \\
\text{Operational saving/month} &= 32{,}500 \times (\textbf{Rs.}\ 800 - \textbf{Rs.}\ 25) = \textbf{Rs. 2.52 crore} \\
\text{Total monthly benefit} &= \approx \textbf{Rs. 48 crore} \\
\text{Annual benefit} &= \approx \textbf{Rs. 576 crore}
\end{aligned}$$

CLV Multiplier — The Number Business Judges Rarely See

An education loan customer is a 25-year-old at the start of their financial life. Poonawalla's next product for them is a personal loan in year 3, a car loan in year 5, a home loan in year 8. The CLV of a recovered education loan applicant, across all product lines at conservative cross-sell rates, is approximately **Rs.** 2.8 lakh in net revenue over 15 years. Recovering 11,375 such applicants per month creates a CLV pipeline of **Rs.** 3,185 crore over 15 years — from drop-offs that currently go to zero.

6.3 The Rs. 515 Billion Market Context

India's digital lending market is projected to reach USD 515 billion by 2030 (NUALS/EY). Education loans are among the fastest-growing segments as tier 2/3 city students access higher education in rising numbers. Loan Wizard's vernacular language support (Sarvam AI) makes it natively deployable to this underserved segment — where Poonawalla's growth mandate is explicitly focused, and where no competitor has built a comparable system.

7. We Thought of Everything: Assumptions, Constraints & Edge Cases

7.1 Stated Assumptions

1. Poonawalla has existing bureau API access (CIBIL/Experian) — the CUS uses this as its highest-weight signal.
2. The applicant has a smartphone with front camera and minimum 2G connectivity — minimum bar for 72% of Indian smartphone users.
3. NIRF data is used as a *proxy* for placement probability, not as a definitive claim. Weights are configurable by Poonawalla's credit team.
4. All weights in the CUS are initial values for calibration. Post-deployment, they are tuned quarterly against loan performance data.

7.2 Edge Cases We Have Designed For

Edge Case	Our Handling	Impact
Low-bandwidth rural connection	Adaptive bitrate video; falls back to audio-only STT mode	Maintains data capture
Applicant drops off mid-AI-session	Async session save + full context preservation; resumes from conversational state	No restart required
Vernacular code-switching	Sarvam AI IndicConformer handles Hindi-English mixing natively	Accurate STT
Face not visible / poor lighting	Agent provides live guidance; 3 retry attempts before human escalation	KYC integrity
Aadhaar/DigiLocker unavailable	Fallback to manual document upload with fraud flag increase	Compliant fallback
Applicant is a co-applicant scenario	Agent detects "co-applicant" mention via NLP and triggers co-sign verification flow	Correct application type
Multiple simultaneous drop-off campaigns	Session isolation via Kafka partition key on applicant ID	No cross-contamination
Deepfake video attempt	Liveness detection (eye-blink, head-turn challenge) + frame consistency analysis	V-CIP compliance

Table 7: Edge case handling

8. Built to Survive: Failure Handling & Resilience Design

8.1 Graceful Degradation Hierarchy

The system is designed to degrade gracefully rather than fail catastrophically. Each layer has a defined fallback:

1. **Video call drops mid-session:** System saves full session state to Pinecone vector store. On resume (same or new device), agent reloads context and continues from exact conversational position.
2. **LLM inference latency spike:** Pre-computed risk band from bureau data provides minimum-viable offer while LLM catches up. User sees progressive reveal.
3. **STT service failure:** Falls back to structured text-input form pre-populated with auto-fill data from prior capture. Session is not lost.
4. **Bureau API timeout:** CUS computed with $w_1 = 0$ and remaining weights renormalised. Offer generated at conservative Tier C, flagged for human review.
5. **Fraud score inconclusive:** System escalates to human loan officer review queue with full XAI confidence report, rather than making an automated decision.

8.2 Security Architecture

- All video sessions encrypted end-to-end (DTLS-SRTP via WebRTC)
- V-CIP infrastructure blocks connections from non-India IP addresses and VPN/proxy exits (RBI mandate)
- All data stored within India (RBI data localisation requirement, DLG 2022 Para 10–11)
- AES-256 encryption at rest for video recordings and transcripts
- CERT-In compliant cybersecurity framework for NBFC sector

- Applicant data deletion available on request (DPDP Act 2023 right to erasure)

9. Compliant by Design: RBI V-CIP, DLG 2022 & DPDP Readiness

Why Compliance Is a Competitive Moat, Not a Checkbox
Most hackathon submissions treat compliance as a footnote. We treat it as a **product feature**. A system that is not RBI-compliant cannot be deployed by Poonawalla, regardless of technical sophistication.
Every requirement below is mapped to a concrete system component.

Regulation	Requirement	Our Implementation
RBI V-CIP (2020)	Real-time synchronous session (no pre-recorded video)	WebRTC live session with liveness challenge (blink + head turn)
	Customer geo-tagged within India	GPS capture + IP geo-fencing
	Facial recognition + liveness check	FaceMesh + custom liveness + age estimation
	Secure video storage by RE	Encrypted object storage under RE-owned domain
RBI DLG 2022	KFS delivered before loan execution	Auto-generated KFS, read aloud, digitally signed (Leegality)
	APR disclosed to borrower	Included in KFS + verbally communicated during call
	Cooling-off period (min. 3 days for $>$ 7-day loans)	Timestamp logging + automated 72-hour cancellation window
	No disbursal to third-party accounts	Bank account verification via Penny Drop API
	Adverse Action Notice (AAN)	Auto-generated from CUS decline outputs
DPDP Act 2023	Data minimisation	Only camera, microphone, GPS with explicit consent
	Reporting to CICs	Integration with CIBIL API post-disbursal
	Explicit consent with audit trail	Recorded verbal consent + SMS/email confirmation
	Right to erasure	Data deletion endpoint in audit system
	Data localisation	AWS/GCP India region (Mumbai), no cross-border transfer
	Purpose limitation	Data use restricted to underwriting, declared per step

Table 8: *Regulatory compliance mapped to system implementation*

Note on RBI Digital Lending Directions (2025):
The RBI issued updated Digital Lending Directions on May 8, 2025, superseding the 2022 guidelines. Our architecture is designed for **regulatory adaptability**, using configurable compliance modules instead of hard-coded rules.

10. What's Next: The Roadmap Beyond the Hackathon

10.1 Phase 1 — Hackathon Prototype (Current)

- Functional demo: BIS-scored entry, video session with fraud triangulation, CUS computation, offer generation
- Simulated bureau data + NIRF public dataset integration
- Basic officer dashboard with confidence bands
- RBI compliance layer (KFS, consent trail, adverse action notice)

10.2 Phase 2 — Pilot Deployment (Months 1–3)

- Live CIBIL/Experian bureau API integration
- DigiLocker API for real-time Aadhaar XML and PAN verification
- Sarvam AI IndicConformer STT in production (all 6 target languages)
- Thompson Sampling bandit activated — begins accumulating Beta distribution data
- PSI monitoring dashboard live — weekly model health reports
- 500-session controlled pilot with Poonawalla education loan drop-off cohort

10.3 Phase 3 — Full Production (Months 3–12)

- GNN fraud ring detection model trained on Poonawalla's application graph
- Thompson Sampling reaches convergence ($\sim$10,000 sessions); question variants optimised
- CUS weights recalibrated against first cohort's loan performance data (3-month repayment signal)
- Expansion to personal loans, MSME loans — same architecture, different policy configurations
- API-first design enables white-labelling to other NBFCs — potential SaaS revenue stream

10.4 The Moat That Compounds

Loan Wizard is not a static system. Thompson Sampling improves question phrasing over 10,000 sessions. The GNN fraud model densifies as application graph grows. CUS weights self-calibrate against actual repayment data. Session memory makes every returning applicant's experience better than their last. The system's competitive advantage *compounds* with usage — this is a data flywheel, not a feature list.

10.5 Integration Opportunities

- **Perfios:** Bank statement cashflow analysis as an additional CUS signal — directly addresses the 34% thin-file misclassification gap they identified
- **Account Aggregator (AA) Framework:** RBI's AA framework enables consent-based financial data sharing; Loan Wizard's consent architecture is AA-compatible by design
- **ONDC Credit:** Open Network for Digital Commerce credit integration for education-related merchant lending

Loan Wizard is not a demo. It is a deployable product blueprint.

Every formula is calibratable. Every compliance item is implemented. Every architectural decision has a production rationale. The market exists, the technology is proven, and the regulatory framework is navigated. What Poonawalla has is a drop-off problem. What we have built is the solution.

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